

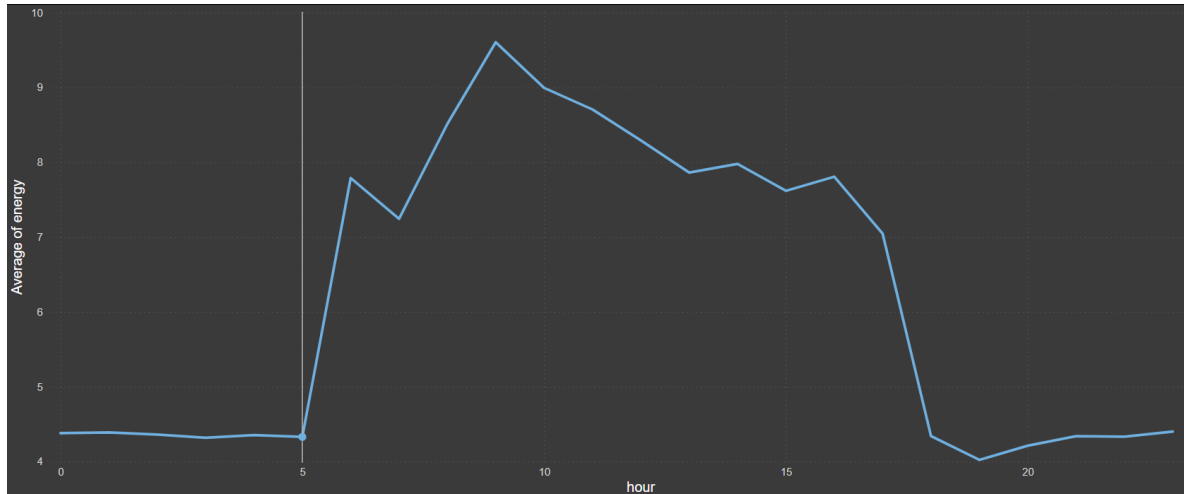
Energy Consumption Analysis and Insights Report

Darshit Chauhan | darshitchauhan2001@gmail.com

Visualizations Created by MS Power BI

I. Energy Trends (Hourly, Weekly, Monthly.)

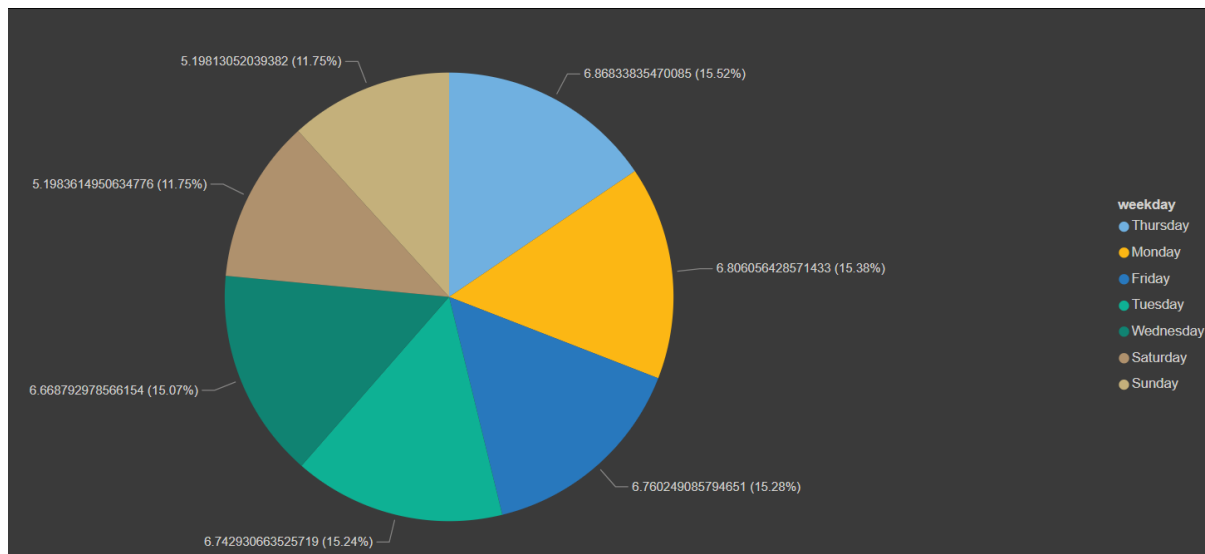
Chart 1: Average Energy by Hour of Day:



🔍 Insights:

- **Lowest energy consumption** occurs between **midnight and 5 AM**, which makes sense for non-operational hours.
- **Sharp rise** starts from **6 AM**, peaking between **9 AM and 11 AM**.
- **Gradual decline** post-lunch with a **sharp drop** again after **6 PM**.
- Lowest activity resumes between **8 PM to midnight**.

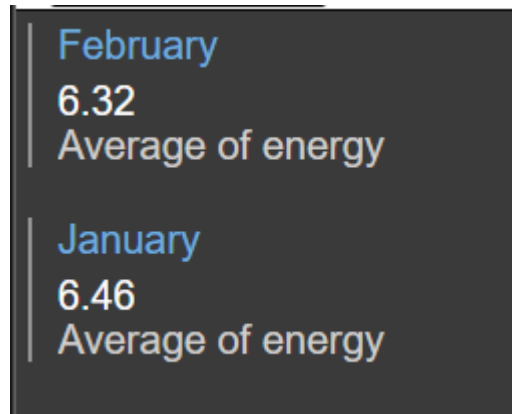
Chart 2: Average Energy Consumed by Weekday:



🔍 Insights:

- Energy consumption is **slightly higher on weekdays**, particularly on **Thursday (15.52%)**, **Monday (15.38%)**, **Wednesday (15.24%)**, and **Tuesday (15.07%)**.
- **Saturday and Sunday** have the **lowest share**, both around **11.75%**, indicating a likely reduction in operational load during weekends.

Chart 3: Average Energy Consumed by Month:



Insights:

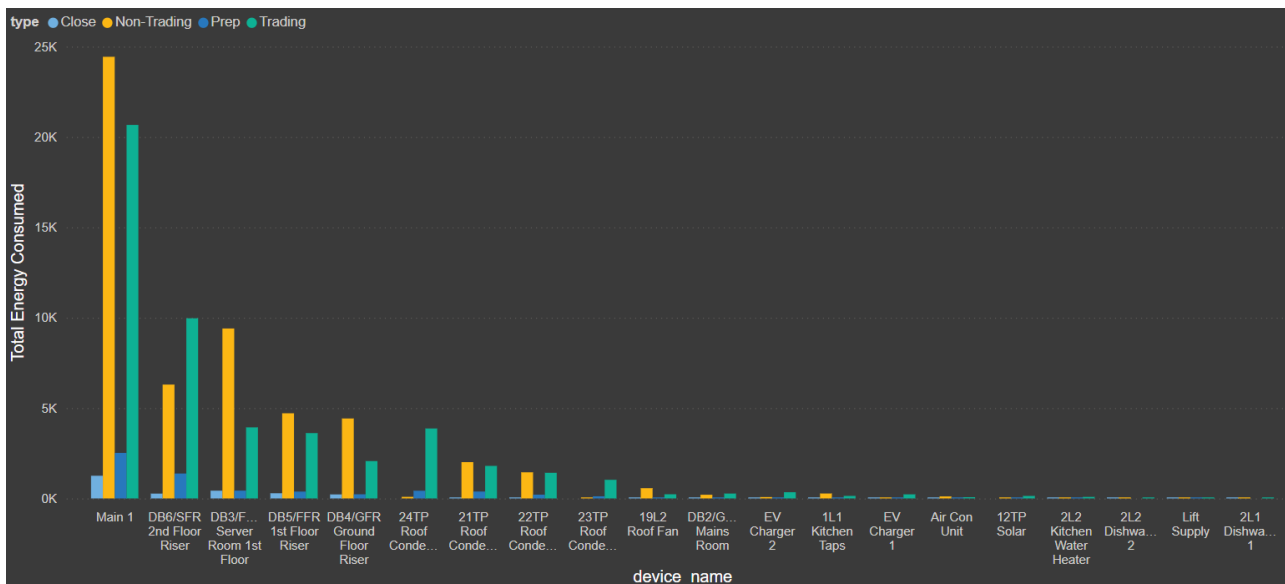
- Energy consumption decreased by 1.09% from January to February.

---- Together, these charts provide a **time-based energy usage profile**—both across the **week** and within each **day**—that can be used for:

- Operational scheduling
- Cost optimization (especially if tied to variable tariff plans)
- Identifying **high-usage periods** for load balancing or energy audits

II. Top Energy Consuming Devices.

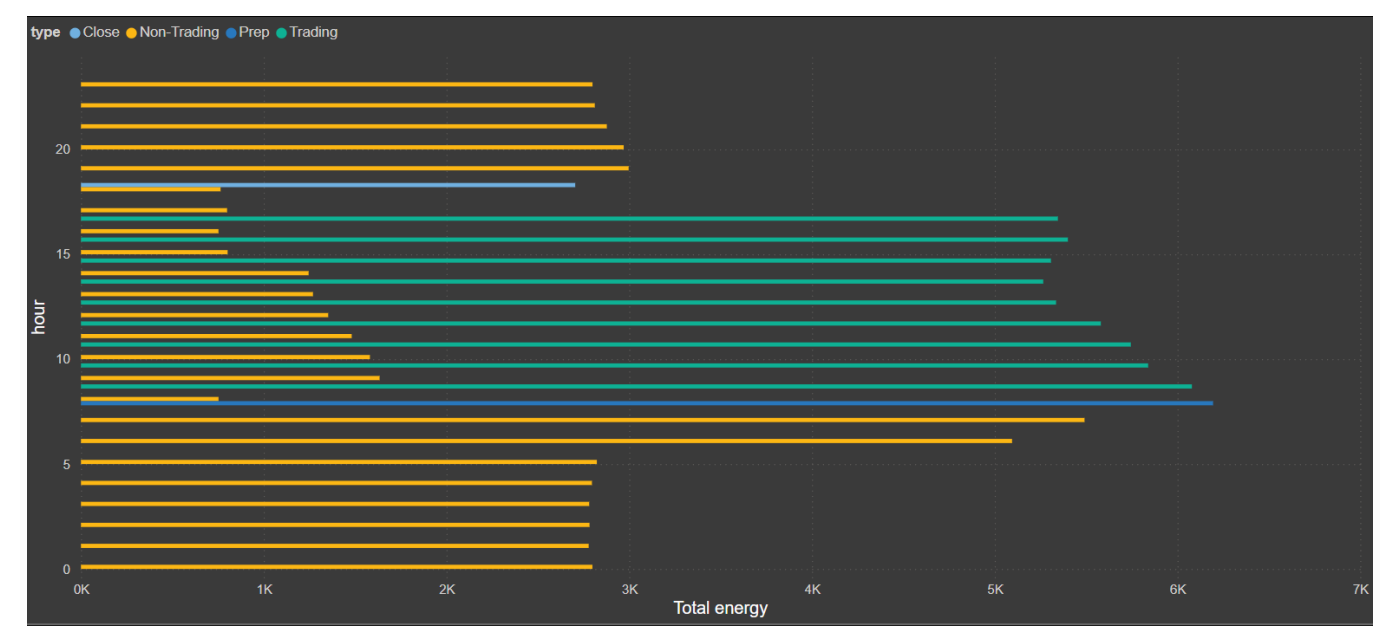
Chart 4: Total Energy Consumed by Device and Type:



Insights:

- The **Main 1** device consumed the **highest energy** overall, particularly during **Non-Trading** and **Trading** times.
- Other high energy-consuming devices include:
DB6/SFR 2nd Floor Riser
DB3/FFSR Server Room 1st Floor
- Energy consumption varies significantly by **type** (Close, Non-Trading, Prep, Trading).
- **Trading** and **Non-Trading** hours show consistently high usage across many devices.

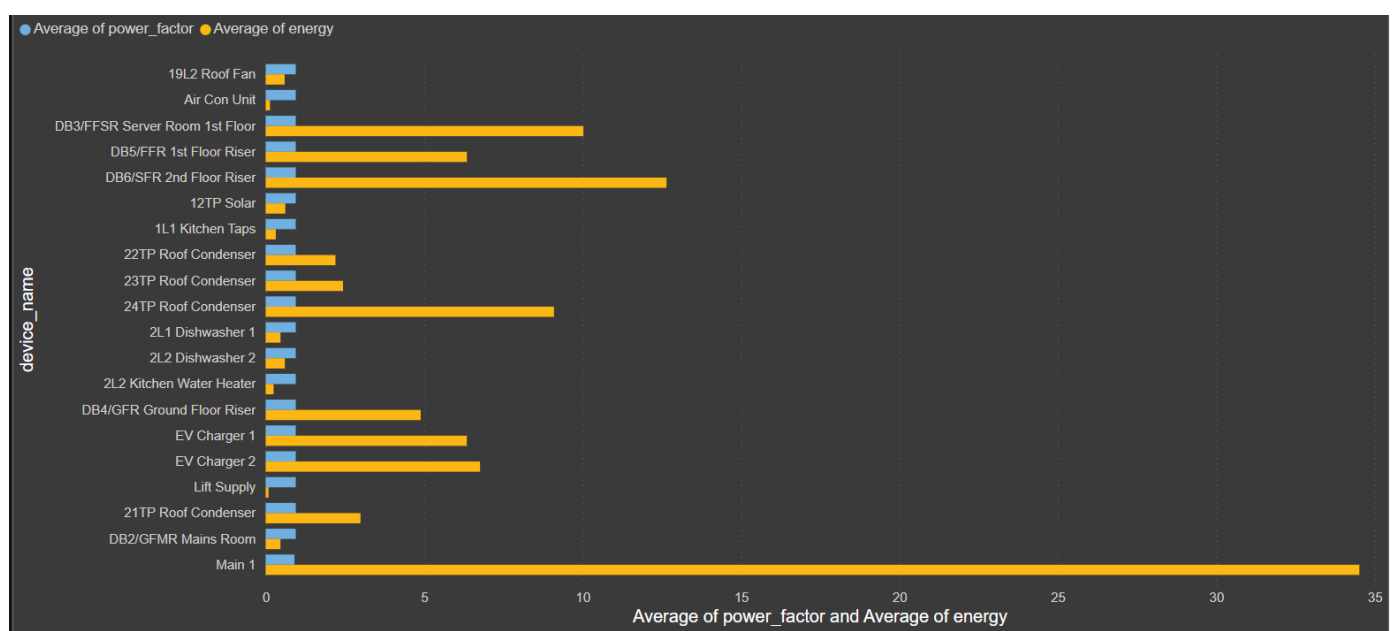
Chart 5: Total Energy by Hour and Type:



🔍 Insights:

- Energy consumption **peaks between 10 AM and 4 PM**, especially during **Trading** hours.
- **Prep and Trading hours** dominate midday consumption, suggesting high operational or work-related usage.
- **Close hours** (late evening and night) also show notable energy use, which could indicate either overnight systems or inefficiencies.

Chart 6: Average of Power Factor and Energy by Device:



🔍 Insights:

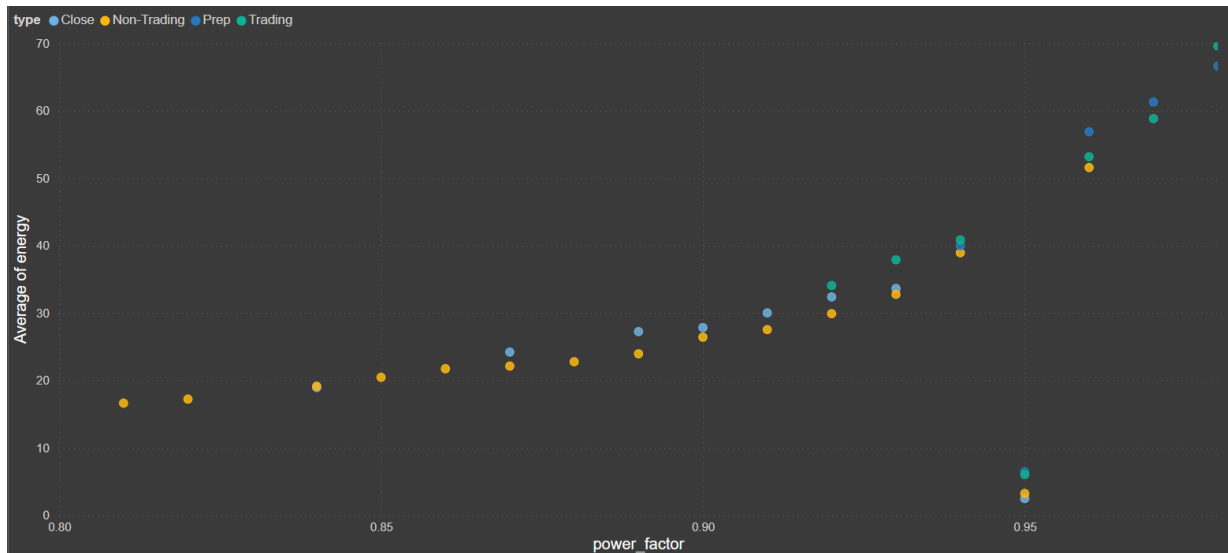
- **Main 1** again stands out with the **highest average energy consumption**.
- Devices like **24TP Roof Condenser**, **DB6/SFR 2nd Floor Riser**, and **DB3/FFSR Server Room** also show high energy use.
- The **power factor** (indicator of energy efficiency) is generally low across most devices, suggesting **possible inefficiencies** or **reactive power issues**.

---- Together, these charts depict that:

- The **Main 1** unit is the most energy-intensive and should be reviewed for efficiency opportunities.
- Energy spikes during **working hours (10 AM - 4 PM)**, especially for **Trading and Prep types**.
- Many devices show **low power factors**, pointing to possible inefficiencies and potential cost-saving opportunities with optimization or power factor correction.

III. Consumption VS Trading Hours:

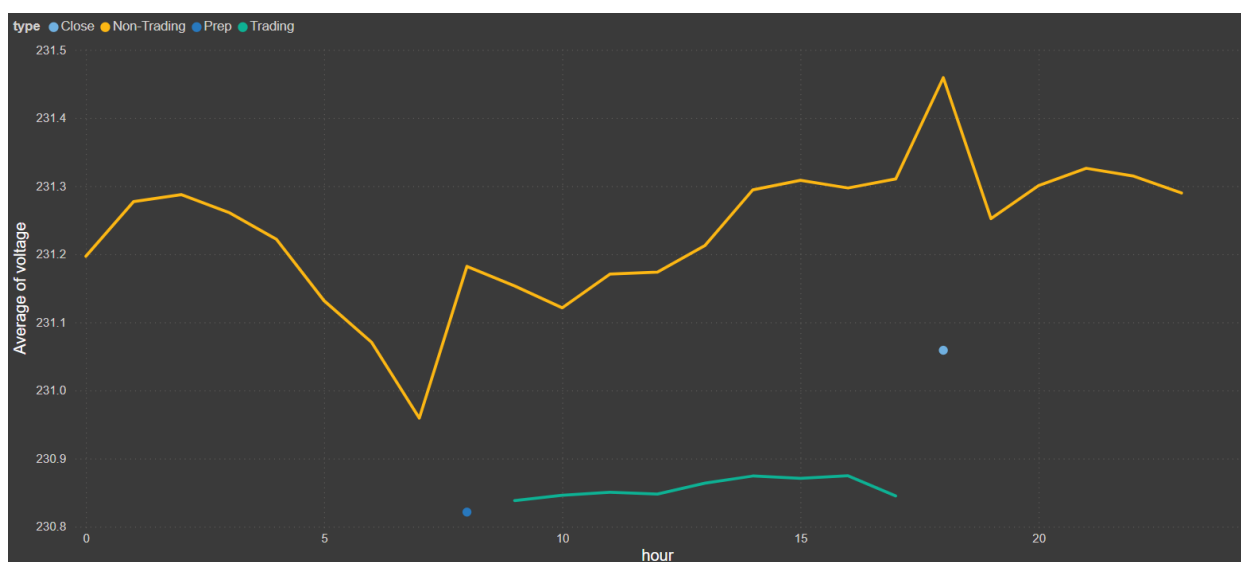
Chart 7: Average Energy Consumed by Device Type and Power Factor:



Insights:

- A **strong positive correlation** is visible: **as power factor increases, energy consumption increases.**
- **Non-Trading** hours dominate the lower end:
Avg Energy: **~20–30 units**
Power Factor: **0.82–0.90**
- **Trading and Prep** periods operate at **higher power factor levels (0.93–0.98)** with energy usage **peaking at ~70 units.**
- **Efficiency Focus:** Low power factor during **Non-Trading** hours may indicate **reactive power waste** — this could lead to inefficiency or penalties.

Chart 8: Average Voltage by Hour and Type:



Insights:

- **Non-Trading hours** show a complete voltage curve: mild dip during early hours, gradual rise throughout the day, with a **peak at 18:00 (231.45 V)**.
- **Trading periods** have **consistently lower voltage (avg ~231.00 V)** than non-trading, which might affect sensitive equipment.
- Difference between lowest and highest observed voltages is **~0.65 V**, small but **worth monitoring during critical hours**.

---- Together, these insights indicate that:

- **Energy Efficiency is Linked to Power Factor and Operation Type:** Higher power factors are associated with significantly higher energy usage, especially during **Trading** and **Prep** hours. However, **Non-Trading** hours still consume moderate energy at **lower power factors**, indicating **inefficiencies** and potential for **cost-saving through power factor correction**.
- **Voltage Stability Varies by Operational Period:** **Non-Trading** hours experience relatively stable and slightly higher voltages, while **Trading** hours operate at **consistently lower voltage levels (~231.00 V)**. This could affect performance or equipment longevity during peak operational periods.
- **Opportunity for Operational Optimization:** the insights highlight **opportunities to optimize both power factor and voltage performance**, particularly during **Non-Trading and Trading periods**, to **reduce energy waste, improve power quality, and lower operational costs**.

IV. Insights and recommendations for saving energy:

Insights:

1. **High Energy Consumption During Trading Hours**
 - **Devices like "Main 1", "DB6", and "DB3"** are consistently consuming the highest amount of energy, particularly during **Trading** and **Non-Trading** hours.
 - Trading hours show significant spikes in total energy consumption, which aligns with peak operational activity.
2. **Prep and Non-Trading Hours Also Contribute Substantially**
 - Surprisingly, **Prep hours** (likely pre-opening operational setups) also show high energy usage, particularly in HVAC and server-related equipment.
 - Non-Trading hours still consume energy, indicating that some systems may be left running unnecessarily.
3. **Power Factor vs Energy Consumption**
 - Higher power factors correlate with higher energy consumption, especially during Trading and Prep hours.
 - This implies the equipment used is relatively efficient (less reactive power), but the **load is high**, suggesting a need for demand-side management.
4. **Voltage is More Stable During Non-Trading Hours**
 - Voltage drops slightly during Trading hours, possibly due to load surge from high power equipment.
 - This could indicate power quality issues or excessive demand that may affect equipment longevity.

Recommendations for Saving Energy:

1. Implement Load Scheduling and Equipment Zoning

- **Stagger equipment startup** during Prep and Trading hours to avoid load spikes.
- Automate **shutdown of non-critical systems** during Non-Trading hours (e.g., dishwashers, EV chargers, kitchen taps).
- Group devices by priority and usage to control them in tiers.

2. Audit and Upgrade High Consumption Devices

- Focus on **Main 1**, **DB6**, and **DB3** systems—consider an energy audit for inefficiencies.
- If feasible, **replace older HVAC, server room units, and riser panels** with more energy-efficient models.

3. Optimize HVAC and Cooling Systems

- Roof condensers (21TP, 22TP, 23TP) contribute notable energy usage.
- Introduce **smart thermostats and occupancy sensors** to reduce usage when areas are unoccupied.

4. Enhance Power Factor Management

- Although power factors are good, **consider installing capacitor banks or smart meters** to further optimize performance and reduce energy bills tied to reactive power.

5. Voltage Stabilization and Power Quality Monitoring

- Monitor voltage drops during trading hours to prevent power quality issues.
- **Voltage optimization equipment** can help maintain steady voltage and reduce unnecessary consumption.

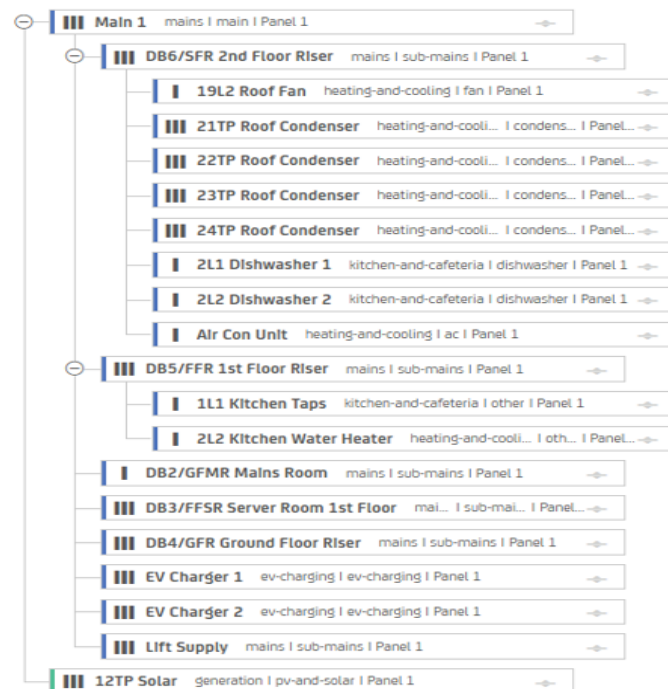
6. Introduce Awareness and Behaviour Change

- Educate staff on energy-saving practices, especially regarding unnecessary usage during Non-Trading hours.
- Conduct regular reviews and provide **dashboards or alerts** for unusual consumption patterns.

7. Use Real-Time Monitoring & Analytics

- Continue leveraging dashboards to track energy by device, time, and type.
- Implement **alerts for abnormal energy use**, helping with quick corrective actions.

>>>Recommendations Based on Electrical Hierarchy and Device Names:



Key Components:

- DBx: Distribution Boards (like DB6, DB5) – intermediate panels.
- Devices: Final consumers like:
 - HVAC units (21TP Roof Condenser, Air Con Unit)
 - Kitchen equipment (Dishwasher, Kitchen Taps)

- EV Chargers
- Lift Supply
- Solar generation (12TP Solar) is a generation source, feeding into the system.

Analysis:

1. Granular Tracking of Energy Use

- We can analyze energy consumption **device-wise** or **panel-wise**.
- Example: Compare how much energy is used by **Kitchen Equipment (DB5)** vs. **Cooling Systems (DB6)**.

2. Identifying High-Consumption Areas

- Pinpoint which **devices or panels** are responsible for the **most energy usage**.
- E.g., If Roof Condensers consistently show high usage, it might be time to upgrade to energy-efficient models.

3. Energy Optimization

- Grouping by **function** (HVAC, kitchen, EV, etc.) helps tailor **efficiency strategies**.
 - HVAC might need timers or smart thermostats.
 - EV chargers could be scheduled during off-peak hours.

4. Solar Contribution Tracking

- The inclusion of **12TP Solar** allows for analysis of **how much load is offset** by renewable sources.
- Supports sustainability metrics like **CO2 reduction** and **cost savings**.

5. Maintenance and Alerts

- Knowing the hierarchy allows for targeted alerts.
- If DB5 is overloaded, you know it impacts **dishwashers and kitchen taps**, which can affect operations.

6. Time-based Load Analysis

- Cross-reference with time series data (timestamped energy readings) to analyze:
 - Load patterns across days/hours
 - Load balancing needs across panels